



2IA Practice Problems III

Groups



Practice Problems Chapter 3

1. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**
 - (a) If H and K are two subgroups of a group G , then their union $H \cup K$ is also subgroup of G .
 - (b) The cyclic group $\mathbb{Z}_n$ has exactly n generators.
 - (c) Every abelian group is cyclic.
 - (d) The intersection of any two subgroups of a group G can never be the empty set.
 - (e) $\mathbb{Z}_2$ is a subgroup of $\mathbb{Z}_4$.
 - (f) Every two cyclic groups of the same order are isomorphic.
 - (g) There exists group G that is isomorphic to a proper subgroup of itself.
2. Consider an arbitrary group G and let $a, b \in G$.
 - (a) Use induction to prove that $b(ab)^{n-1}a = (ba)^n$ for all $n \geq 1$.
 - (b) Suppose now that $(ab)^m = 1$ for some $m \geq 1$. Use (a) to show that $(ba)^m = 1$.
3. Let a and b elements in a group G such that they commute, i.e., $ab = ba$. Prove that a^{-1} and b^{-1} commute.
4. Let G be a group and $a, b \in G$ arbitrary elements.
 - (a) (10 marks) Prove (by induction on n) that
$$(aba^{-1})^n = ab^n a^{-1}, \text{ for every integer } n \geq 1.$$
 - (b) (5 marks) Prove that $(aba^{-1})^n = ab^n a^{-1}$ for all integers n .
5. Let H and K be subgroups of a group G . Consider the subset HK of G defined as follows:

$$HK = \{hk \mid h \in H, k \in K\}$$

Prove that HK is a subgroup of G if and only if $KH \subseteq HK$; notice that KH is defined exactly in the same way: $KH = \{kh \mid h \in H, k \in K\}$.

6. Let G be a group and $g \in G$. Prove that $\langle g \rangle = \langle g^{-1} \rangle$.

Hint: Show that $\text{o}(g) = \text{o}(g^{-1})$.

7. Find all generators of

(a) $G = \langle g \rangle$, where $\text{o}(g) = \infty$.

(b) $\mathbb{Z}_{16}$.

(c) $G = \langle g \rangle$, if $\text{o}(g) = 5$.

8. (a) Determine all the elements of the multiplicative groups $\mathbb{Z}_{14}^*$ and $\mathbb{Z}_{18}^*$.
(b) Prove that $\mathbb{Z}_{14}^*$ and $\mathbb{Z}_{18}^*$ are both cyclic groups of order 6. Conclude that there are isomorphic.

Recall that $\mathbb{Z}_n^*$ denotes the subset of $\mathbb{Z}_n$ consisting on the elements that have a multiplicative inverse.

9. Consider the additive cyclic group $\mathbb{Z}_{48}$.

(a) Determine all the subgroups of $\mathbb{Z}_{48}$; list all the elements of each subgroup.

(b) Draw the lattice diagram of all the subgroups $\mathbb{Z}_{48}$.